Thm. Let G be a finite group, and let $g_1, g_2, \cdots, g_n$ be representatives of the distinct Conjugacy classes of G, not contained in the Center $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|.$$

Note: $C_G(g_i) = N_G(g_i)$, and so $|G : N_G(g_i)| = C_{g_i} = \{g g_i g^{-1} \mid g \in G\}$.

Thm. If p is a prime number and G is group of order p^α $(\alpha \geq 1)$, then G have a non trivial Center.

Proof. By the class equation,

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|.$$

But, since $C_G(g_i)$ is a sub group of G and $C_G(g_i) \neq G$, because $g_i \notin Z(G)$. Therefore, by the Lagrange's Thm, p divides $|G : C_G(g_i)|$ for each i, so p also divides $|Z(G)|$. Thus, it cannot be trivial.

Corollary. If $|G| = p^2$ for some prime p, then P is abelian